

Dirichlet Characters

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Abstract

We give an exposition of Dirichlet characters with a view towards Langlands reciprocity. We assume the content of a graduate course in abstract algebra (groups, rings, fields, Galois theory, etc.). In an actual talk, it may be profitable to start by stating the main result Corollary 27 as quickly as possible, rearranging the exposition to cover Section 2 before Section 1. Section 3 should only be covered if there is ample time at the end. This note is based on the blog posts [Elb25a; Elb25b].

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1 The Automorphic Side

In this section, we relate Dirichlet characters to objects appearing on the automorphic side of the Langlands program.

1.1 Classical Constructions

Let's recall the classical definition of a Dirichlet character.

Definition 1 (Dirichlet character). Fix a positive integer N . Then a *Dirichlet character* $(\bmod N)$ is a group homomorphism $\chi: (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$.

Example 2. Here is a Dirichlet character $\chi: (\mathbb{Z}/5\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$.

n	1	2	3	4
$\chi(n)$	1	i	$-i$	-1

Example 3. Fix a prime p . It turns out that the group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $(p-1)$. Thus, for a choice of generator $g \in (\mathbb{Z}/p\mathbb{Z})^\times$, we see that a Dirichlet character $\chi: (\mathbb{Z}/p\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ is uniquely determined by $\chi(g)$, which must be a $(p-1)$ st root of unity.

Example 4. Fix an odd prime p . Because $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of even order, it admits a unique subgroup of index two, so there is a unique surjective homomorphism

$$\chi_p: (\mathbb{Z}/p\mathbb{Z})^\times \rightarrow \{\pm 1\}.$$

This is called the quadratic Dirichlet character $(\text{mod } p)$.

Roughly speaking, we are interested in enumerating Dirichlet characters. To this end, this definition has a defect: from a given Dirichlet character $(\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$, one can lift it to a Dirichlet character

$$(\mathbb{Z}/NM\mathbb{Z})^\times \rightarrow (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times.$$

This new Dirichlet character does not add any new information, so we would like to remove them from our discussion.

Definition 5 (conductor). Fix a Dirichlet character $\chi: (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$. Then the *conductor* of χ is the least divisor d of N such that χ factors through $(\mathbb{Z}/d\mathbb{Z})^\times$. The character χ is *primitive* if and only if its conductor is N .

Remark 6. Here is an annoying check: if χ factors through $(\mathbb{Z}/M\mathbb{Z})^\times$, then M is divisible by the conductor d . Indeed, χ factors through both M and d implies that

$$\{n \in (\mathbb{Z}/N\mathbb{Z})^\times : n \equiv 1 \pmod{M} \text{ or } n \equiv 1 \pmod{d}\} \subseteq \ker \chi.$$

Expanding out everything into prime-powers via the Chinese remainder theorem reveals that any n for which $n \equiv 1 \pmod{\gcd(M, d)}$ is also in $\ker \chi$, so χ factors through $\mathbb{Z}/\gcd(M, d)\mathbb{Z}$. Thus, $d = \gcd(M, d)$ by minimality of d , so $d \mid M$.

1.2 Working at Infinite Level

It will be convenient to gather together all Dirichlet characters together. In our current formulation, we have some difficulty with this because the domains of our Dirichlet characters vary. If we wanted to create a uniform domain, we could try using

$$\prod_{N \geq 1} (\mathbb{Z}/N\mathbb{Z})^\times,$$

but this turns out to be too big for applications. For example, this infinite product does not remember any of the “arithmetic” information about the groups $(\mathbb{Z}/N\mathbb{Z})^\times$, which lies in divisibility. We are thus motivated to build the following object.

Notation 7. Define

$$\widehat{\mathbb{Z}}^\times := \left\{ (a_N)_N \in \prod_{N \geq 1} (\mathbb{Z}/N\mathbb{Z})^\times : a_{dN} \equiv a_N \pmod{N} \text{ for any } d, N \geq 1 \right\}.$$

We give $\widehat{\mathbb{Z}}^\times$ the subspace topology from the infinite product $\prod_N (\mathbb{Z}/N\mathbb{Z})^\times$, where each finite group $(\mathbb{Z}/N\mathbb{Z})^\times$ is discrete.

Remark 8. The subgroup $\widehat{\mathbb{Z}}^\times$ is a closed and hence compact subset of $\prod_N (\mathbb{Z}/N\mathbb{Z})^\times$. In short, $\widehat{\mathbb{Z}}^\times$ is cut out by the continuous functions $f_{N',d'} : \prod_N (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow (\mathbb{Z}/N'\mathbb{Z})^\times$ defined by

$$f_{N',d'}((a_N)_N) := a_{N'd'} - a_{N'}.$$

Remark 9. One can define $\widehat{\mathbb{Z}}$ as the ring

$$\widehat{\mathbb{Z}} := \left\{ (a_N)_N \in \prod_{N \geq 1} (\mathbb{Z}/N\mathbb{Z}) : a_{dN} \equiv a_N \pmod{N} \text{ for any } d, N \geq 1 \right\}.$$

It is not hard to check that $\widehat{\mathbb{Z}}^\times$ is the group of units of $\widehat{\mathbb{Z}}$. One approach by “pure thought” is to show the functor $(-)^\times : \text{Ring} \rightarrow \text{AbGrp}$ preserves limits because it is right adjoint to the functor $A \mapsto \mathbb{Z}[A]$.

Remark 10. From the perspective of the Langlands program, $\widehat{\mathbb{Z}}^\times$ is still not the correct object. Instead, one should study $\mathbb{Q}^\times \backslash \mathbb{A}_\mathbb{Q}^\times$, where $\mathbb{A}_\mathbb{Q}^\times$ is the group of ideles. However, defining the ideles is slightly beyond the scope of this note, so we will say nothing about them other than to say that $\mathbb{Q}^\times \backslash \mathbb{A}_\mathbb{Q}^\times / \mathbb{R}^+ \cong \widehat{\mathbb{Z}}^\times$.

Here is the main result of this section.

Proposition 11. The set of continuous homomorphisms $\widehat{\mathbb{Z}}^\times \rightarrow \mathbb{C}^\times$ is in canonical bijection with the set of primitive Dirichlet characters.

Most of the work of the proof is carried out in the following claim.

Lemma 12. Any continuous homomorphism $\widehat{\mathbb{Z}}^\times \rightarrow \mathbb{C}^\times$ factors through $(\mathbb{Z}/N\mathbb{Z})^\times$ for some positive integer N .

Proof. This is an instance of the “no small subgroups” argument. Fix a continuous character $\chi : \widehat{\mathbb{Z}}^\times \rightarrow \mathbb{C}^\times$.

1. We claim that it is enough to show that χ contains a nonempty open subset in its kernel. Then the kernel contains a basic open neighborhood of $1 \in \widehat{\mathbb{Z}}^\times$, which we may shrink to look like

$$\widehat{\mathbb{Z}}^\times \cap \prod_{N \in S} \{1\} \times \prod_{N \notin S} (\mathbb{Z}/N\mathbb{Z})^\times,$$

where S is some finite set of positive integers. Thus, χ factors through the image of the projection of $\widehat{\mathbb{Z}}^\times$ to $\prod_{N \in S} (\mathbb{Z}/N\mathbb{Z})^\times$, so χ factors through $(\mathbb{Z} / \prod_{N \in S} N\mathbb{Z})^\times$.

2. Let $V \subseteq \mathbb{C}^\times$ be the open disk of radius one around $1 \in \mathbb{C}^\times$. We claim that V contains no nontrivial subgroup of $\mathbb{C}^\times$. Indeed, choose $z \in V \setminus \{1\}$. On one hand, if $|z| \neq 1$ implies that $|\langle z \rangle|$ is unbounded and not contained in U . On the other hand, if $|z| = 1$, we may write $z = e^{i\theta}$ for some $\theta \in (0, \pi/3) \cup (-\pi/3, 0)$, which implies that $z^2 \notin U$.

3. We are now ready to show that $\ker \chi$ contains an open subset. Well, $\chi^{-1}(V)$ is open and contains 1, so $\chi^{-1}(V)$ contains a basic open neighborhood of 1, which we may shrink to be of the form

$$U := \widehat{\mathbb{Z}}^\times \cap \prod_{N \in S} \{1\} \times \prod_{N \notin S} (\mathbb{Z}/N\mathbb{Z})^\times,$$

where S is some finite set of positive integers. We claim that U will suffice. Indeed, U is a subgroup, so $\chi(U) \subseteq V$ is a subgroup, so $\chi(U)$ is trivial, so $U \subseteq \ker \chi$. ■

Proof of Proposition 11. In one direction, any primitive Dirichlet character $\chi: (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ induces a continuous character

$$\widehat{\mathbb{Z}}^\times \rightarrow (\mathbb{Z}/N\mathbb{Z})^\times \xrightarrow{\chi} (\mathbb{C}^\times).$$

In the other direction, for any continuous homomorphism $\chi: \widehat{\mathbb{Z}}^\times \rightarrow \mathbb{C}^\times$, Lemma 12 implies that χ factors through $(\mathbb{Z}/N\mathbb{Z})^\times$ for some least positive N . It follows that the character $(\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ is primitive by construction of N .

It remains to show that these constructions are inverse. Of course, one can start from a continuous character $\widehat{\mathbb{Z}}^\times \rightarrow \mathbb{C}^\times$, project to the relevant $(\mathbb{Z}/N\mathbb{Z})^\times$, and then lift back to $\widehat{\mathbb{Z}}^\times$ to get the same character. For the other check, if we start with a primitive Dirichlet character $\chi: (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ and lift it to $\widehat{\mathbb{Z}}^\times$, we must show that N equals the smallest positive integer M for which $\widehat{\mathbb{Z}}^\times$ factors through $(\mathbb{Z}/M\mathbb{Z})^\times$. Well, $M \leq N$ by minimality of M , and $N \mid M$ by Remark 6, so $M = N$ follows. ■

2 The Galois Side

In this section, we relate Dirichlet characters to objects appearing on the Galois side of the Langlands program.

2.1 Galois Representations

There comes a time in life where one must ask difficult but fundamental questions. For example, what is the meaning of life? What are we doing here?

Actually, this second question is easier to answer: we are in a talk about algebraic number theory. As for the meaning of algebraic number theory, it basically boils down to the following question.

Question 13. What is the structure of the group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$?

Remark 14. The Galois extension $\mathbb{Q} \subseteq \overline{\mathbb{Q}}$ is infinite, so we must equip the Galois group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ with the Krull topology: view $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ as the subset

$$\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) = \left\{ (\sigma_K)_K \in \prod_{\text{finite Galois } K/\mathbb{Q}} \text{Gal}(K/\mathbb{Q}) : \sigma_L|_K = \sigma_K \text{ if } K \subseteq L \right\}.$$

(Namely, send $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ to the tuple of restrictions $(\sigma|_K)_K$. This map is then an isomorphism.) Thus, we may equip $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ with the subspace topology from the infinite product $\prod_K \text{Gal}(K/\mathbb{Q})$. A similar argument to Remark 8 shows that $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ is a closed and hence compact subset.

It is difficult to study $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ directly. For example, identifying a single element requires an automorphism for each finite Galois extension of $\mathbb{Q}$. Instead, we will take a cue from representation theory and understand $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ via its actions on finite-dimensional complex vector spaces. Even this is too difficult a task for the present note, so we will have to content ourselves with understanding actions on one-dimensional complex vector spaces. This amounts to defining a (continuous) homomorphism

$$\rho: \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}_1(\mathbb{C}).$$

The target is simply $\mathbb{C}^\times$.

Example 15. An argument similar to Lemma 12 can show that any continuous map $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \mathbb{C}^\times$ factors through $\text{Gal}(K/\mathbb{Q})$ for some finite Galois extension K of $\mathbb{Q}$. For example, if $K/\mathbb{Q}$ is a quadratic extension, then we have the continuous homomorphism

$$\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \twoheadrightarrow \text{Gal}(K/\mathbb{Q}) \cong \{\pm 1\} \subseteq \mathbb{C}^\times.$$

By the universal property of the abelianization, it is equivalent to construct a continuous homomorphism $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})^{\text{ab}} \rightarrow \mathbb{C}^\times$. It now turns out that $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})^{\text{ab}}$ can be understood directly.

Definition 16. A Galois extension L/K of fields is *abelian* if and only if its Galois group is abelian.

Notation 17. Let $\mathbb{Q}^{\text{ab}}$ be the largest abelian extension of $\mathbb{Q}$.

Remark 18. To check that $\mathbb{Q}^{\text{ab}}$ is well-defined, we can show that the composite of any two abelian extensions L and K of $\mathbb{Q}$ continues to be abelian. Indeed, because LK is generated by L and K , the product map

$$\text{Gal}(LK/\mathbb{Q}) \rightarrow \text{Gal}(L/\mathbb{Q}) \times \text{Gal}(K/\mathbb{Q})$$

is injective, so the left group is abelian because the right one is.

Remark 19. The same discussion as in Remark 14 applies to $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$.

Lemma 20. Let L be a finite Galois extension of $\mathbb{Q}$, and let $K \subseteq L$ be the largest sub-extension such that $\text{Gal}(K/\mathbb{Q})$ is abelian. Then

$$\text{Gal}(K/\mathbb{Q}) = \text{Gal}(L/\mathbb{Q})^{\text{ab}}.$$

Proof. Note that K is well-defined by the same argument as Remark 18. Now, by construction, $\text{Gal}(L/\mathbb{Q})^{\text{ab}}$ is the largest abelian quotient of $\text{Gal}(L/\mathbb{Q})$. Accordingly, let H be the kernel of $\text{Gal}(L/\mathbb{Q}) \rightarrow \text{Gal}(L/\mathbb{Q})^{\text{ab}}$ (namely, H is the commutator), and H is the smallest subgroup for which $\text{Gal}(L/\mathbb{Q})/H$ is abelian. It follows that L^H is the largest extension of $\mathbb{Q}$ whose Galois group $\text{Gal}(L/\mathbb{Q})/H$ is abelian, so $K = L^H$. The result follows. ■

Proposition 21. The restriction map $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$ induces an isomorphism $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})^{\text{ab}} \rightarrow \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$.

Proof. The idea is to glue together instances of Lemma 20. To avoid annoying considerations of infinite products (though such an argument is certainly possible), we suggest a proof by pure thought: the functor $(-)^{\text{ab}} : \text{Grp} \rightarrow \text{AbGrp}$ is right adjoint to the forgetful functor, so it preserves limits, so

$$\begin{aligned} \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})^{\text{ab}} &= \left(\lim_{\text{finite Galois } K/\mathbb{Q}} \text{Gal}(K/\mathbb{Q}) \right)^{\text{ab}} \\ &= \lim_{\text{finite Galois } K/\mathbb{Q}} \text{Gal}(K/\mathbb{Q})^{\text{ab}} \\ &= \lim_{\text{finite Galois } K/\mathbb{Q}} \text{Gal}(K'/\mathbb{Q}) \end{aligned}$$

where $K' \subseteq K$ is the largest sub-extension with abelian Galois group. (Here we have applied Lemma 20.) This last limit is $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$. ■

Thus, we want to understand $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$.

2.2 The Kronecker–Weber Theorem

It may appear no easier to understand $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$, but $\mathbb{Q}^{\text{ab}}$ turns out to be fairly concrete. For example, it is actually relatively easy to exhibit abelian extensions of $\mathbb{Q}$.

Example 22. For any positive integer N , there is the Galois extension $\mathbb{Q}(\zeta_N)$ of $\mathbb{Q}$, where ζ_N is a primitive N th root of unity (for example, $\zeta_N := \exp(2\pi i/N)$). There is a canonical map

$$\sigma_{\bullet}: (\mathbb{Z}/N\mathbb{Z})^{\times} \rightarrow \text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$$

given by sending k to the automorphism σ_k of $\mathbb{Q}(\zeta_N)$ defined by $\zeta_N \mapsto \zeta_N^k$. One can compute that this map is a well-defined isomorphism.

Remark 23. In the sequel, we will need the following coherence result: if $N \mid N'$, then $\mathbb{Q}(\zeta_N) \subseteq \mathbb{Q}(\zeta_{N'})$ (indeed, $\zeta_N = \zeta_{N'}^{N'/N}$), and the diagram

$$\begin{array}{ccc} (\mathbb{Z}/N'\mathbb{Z})^{\times} & \longrightarrow & \text{Gal}(\mathbb{Q}(\zeta_{N'})/\mathbb{Q}) \\ \downarrow & & \downarrow \\ (\mathbb{Z}/N\mathbb{Z})^{\times} & \longrightarrow & \text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \end{array} \quad \begin{array}{ccc} k & \longmapsto & (\zeta_{N'} \mapsto \zeta_{N'}^k) \\ \downarrow & & \downarrow \\ k & \longmapsto & (\zeta_N \mapsto \zeta_N^k) \end{array}$$

commutes.

This turns out to exhaust all abelian extensions of $\mathbb{Q}$.

Theorem 24 (Kronecker–Weber). Let K be an abelian extension of $\mathbb{Q}$. Then K is contained in $\mathbb{Q}(\zeta_N)$ for some positive integer N .

Proof. We entirely omit this proof. It is a difficult result! On a personal note, I have never read through a direct proof of this theorem without getting overwhelmingly bored. ■

Corollary 25. There is a canonical isomorphism $\hat{\mathbb{Z}}^{\times} \rightarrow \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$.

Proof. As in Remark 14, we have that

$$\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) = \left\{ (\sigma_K)_K \in \prod_{\text{finite abelian } K/\mathbb{Q}} \text{Gal}(K/\mathbb{Q}) : \sigma_L|_K = \sigma_K \text{ if } K \subseteq L \right\}.$$

There is a canonical restriction of tuples map to

$$\left\{ (\sigma_N)_N \in \prod_{\text{even } N} \text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) : \sigma_{N'}|_{\mathbb{Q}(\zeta_N)} = \sigma_N \text{ if } N \mid N' \right\}.$$

Here, we may require N even because $\mathbb{Q}(\zeta_N) = \mathbb{Q}(\zeta_{2N})$ when N is odd. With N even, $\mathbb{Q}(\zeta_N) \subseteq \mathbb{Q}(\zeta_{N'})$ implies that $N \mid N'$: upon passing to the group of roots of unity, we see $\langle \zeta_N \rangle \subseteq \langle \zeta_{N'} \rangle$, so $N \mid N'$. Anyway, the canonical restriction map is surjective: the fiber over $(\sigma_N)_N$ is given by extending the induced automorphism of $\bigcup_N \mathbb{Q}(\zeta_N)$ to an automorphism of $\mathbb{Q}^{\text{ab}}$. Injectivity follows because this extension is unique because $\bigcup_N \mathbb{Q}(\zeta_N) = \mathbb{Q}^{\text{ab}}$ by Theorem 24.

It remains to compute the latter group. Because $\mathbb{Q}(\zeta_N) = \mathbb{Q}(\zeta_{2N})$ when N is odd, we may as well pass to

$$\left\{ (\sigma_N)_N \in \prod_{N \geq 1} \text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) : \sigma_{N'}|_{\mathbb{Q}(\zeta_N)} = \sigma_N \text{ if } N \mid N' \right\}$$

because the odd terms add no new information. (Namely, the natural projection map back to even N is an isomorphism.) We now use Example 22 to compute these Galois groups as

$$\left\{ (a_N)_N \in \prod_{N \geq 1} (\mathbb{Z}/N\mathbb{Z})^\times : \sigma_{a_{N'}}|_{\mathbb{Q}(\zeta_N)} = \sigma_{a_N} \text{ if } N \mid N' \right\}.$$

By Remark 23, this is exactly $\widehat{\mathbb{Z}}^\times$. ■

Remark 26. Define $\mathbb{Q}^{\text{cyc}} := \bigcup_N \mathbb{Q}(\zeta_N)$. Even without using Theorem 24, there is a map $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \rightarrow \text{Gal}(\mathbb{Q}^{\text{cyc}}/\mathbb{Q})$, and there is a canonical isomorphism $\widehat{\mathbb{Z}}^\times \rightarrow \text{Gal}(\mathbb{Q}^{\text{cyc}}/\mathbb{Q})$. The content of Theorem 24 lies in showing that the former restriction map is an isomorphism.

We are now ready to return to Dirichlet characters.

Corollary 27 (Langlands reciprocity). The set of one-dimensional representations of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ are in canonical bijection with the set of primitive Dirichlet characters.

Proof. As discussed previously, a one-dimensional representation has equivalent data to a continuous homomorphism $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \mathbb{C}^\times$, which has equivalent data to a continuous homomorphism $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})^{\text{ab}} \rightarrow \mathbb{C}^\times$. By Proposition 21, this is the same as a continuous homomorphism $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \rightarrow \mathbb{C}^\times$. Lastly, this is the same as a continuous homomorphism $\widehat{\mathbb{Z}}^\times \rightarrow \mathbb{C}^\times$ by Corollary 25. We are now done by Proposition 11. ■

Remark 28. One can attempt to generalize Corollary 27 to other number fields F . The correct statement turns out to be the existence of a continuous homomorphism

$$F^\times \backslash \mathbb{A}_F^\times \rightarrow \text{Gal}(\overline{F}/F)^{\text{ab}}$$

with dense image. (The kernel turns out to be $(F \otimes_{\mathbb{Q}} \mathbb{R})^{\times, \circ}$.) Thus, we can parameterize one-dimensional Galois representations in terms of the “automorphic data.”

Remark 29. One can attempt to classify Galois representations of higher dimension. This is the goal of the Langlands program. In very rough terms, one expects to parameterize n -dimensional representations of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ in terms of certain representations of $\text{GL}_n(\mathbb{Q}) \backslash \text{GL}_n(\mathbb{A}_{\mathbb{Q}})$.

3 Quadratic Characters

The goal of this subsection is to explicate Corollary 27 for the very particular Galois representation in Example 15. At the end, we will stumble into quadratic reciprocity.

3.1 The Characters

In light of Corollary 27, we are interested in computing the primitive Dirichlet character attached to the Galois representation

$$\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \twoheadrightarrow \text{Gal}(K/\mathbb{Q}) \cong \{\pm 1\} \subseteq \mathbb{C}^\times.$$

By Corollary 25 (and perhaps Remark 26), it is equivalent to understand the composite

$$\widehat{\mathbb{Z}}^\times \cong \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \twoheadrightarrow \text{Gal}(K/\mathbb{Q}) \cong \{\pm 1\} \subseteq \mathbb{C}^\times.$$

Let’s give this composite a name.

Notation 30. Fix a quadratic extension K of $\mathbb{Q}$. Then define χ_K to be the continuous character

$$\widehat{\mathbb{Z}}^\times \cong \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \rightarrow \text{Gal}(K/\mathbb{Q}) \cong \{\pm 1\} \subseteq \mathbb{C}^\times.$$

Let's compute a few examples by hand.

Example 31. Note $\sqrt{-1} = \zeta_4$, so $\chi_{\mathbb{Q}(\sqrt{-1})}$ is the character

$$\widehat{\mathbb{Z}}^\times \cong \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \rightarrow \text{Gal}(\mathbb{Q}(\zeta_4)/\mathbb{Q}) \cong (\mathbb{Z}/4\mathbb{Z})^\times \cong \{\pm 1\} \subseteq \mathbb{C}^\times.$$

Unwinding the proof of Corollary 25 reveals that the composite isomorphism $\widehat{\mathbb{Z}}^\times \rightarrow (\mathbb{Z}/4\mathbb{Z})^\times$ is the canonical surjection. Thus, the primitive Dirichlet character attached to $\chi_{\mathbb{Q}(\sqrt{-1})}$ is the Dirichlet character $(\mathbb{Z}/4\mathbb{Z})^\times \cong \{\pm 1\}$.

Example 32. Note $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$ because $\zeta_3 = -\frac{1}{2} + \frac{1}{2}\sqrt{-3}$, so $\chi_{\mathbb{Q}(\sqrt{-3})}$ is the character

$$\widehat{\mathbb{Z}}^\times \cong \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \rightarrow \text{Gal}(\mathbb{Q}(\zeta_3)/\mathbb{Q}) \cong (\mathbb{Z}/3\mathbb{Z})^\times \cong \{\pm 1\} \subseteq \mathbb{C}^\times.$$

As in Example 31, we see that the associated primitive Dirichlet character is $(\mathbb{Z}/3\mathbb{Z})^\times \cong \{\pm 1\}$.

To go beyond examples where $\mathbb{Q}(\zeta_N)$ is quadratic, we will need to make Theorem 24 explicit.

Lemma 33. Fix a quadratic extension K of $\mathbb{Q}$. Then $\chi_K: \widehat{\mathbb{Z}}^\times \rightarrow \mathbb{C}^\times$ factors through $(\mathbb{Z}/N\mathbb{Z})^\times$ if and only if $K \subseteq \mathbb{Q}(\zeta_N)$.

Proof. If $K \subseteq \mathbb{Q}(\zeta_N)$, then $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \rightarrow \text{Gal}(K/\mathbb{Q})$ factors through $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \rightarrow \text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$. We conclude that χ_K factors through the isomorphic map $\widehat{\mathbb{Z}}^\times \rightarrow (\mathbb{Z}/N\mathbb{Z})^\times$.

Conversely, if χ_K factors through $\widehat{\mathbb{Z}}^\times \rightarrow (\mathbb{Z}/N\mathbb{Z})^\times$, then $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \rightarrow \text{Gal}(K/\mathbb{Q})$ factors through the isomorphic map $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \rightarrow \text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$. It follows that $\text{Gal}(\mathbb{Q}^{\text{ab}}/K)$ contains $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}(\zeta_N))$, so $K \subseteq \mathbb{Q}(\zeta_N)$ by Galois theory. ■

Example 34. Note that $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$, so $\chi_{\mathbb{Q}(\sqrt{2})}$ factors through $(\mathbb{Z}/8\mathbb{Z})^\times$. In fact, because $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$, we can see that $\sqrt{2}$ is fixed by σ_1 and σ_7 , but σ_3 and σ_5 send $\sqrt{2}$ to $\zeta_8^3 + \zeta_8^{-3} = -\sqrt{2}$. Thus, $\chi_{\mathbb{Q}(\sqrt{2})}$ is associated to the primitive Dirichlet character (mod 8) given as follows.

n	1	3	5	7
$\chi_{\mathbb{Q}(\sqrt{2})}$	+1	-1	-1	+1

Example 35. Note that $\sqrt{-2} = \zeta_8 + \zeta_8^3$, so $\chi_{\mathbb{Q}(\sqrt{-2})}$ factors through $(\mathbb{Z}/8\mathbb{Z})^\times$. Arguing as in Example 34, $\chi_{\mathbb{Q}(\sqrt{-2})}$ is associated to the primitive Dirichlet character (mod 8) given as follows.

n	1	3	5	7
$\chi_{\mathbb{Q}(\sqrt{-2})}$	+1	+1	-1	-1

3.2 Some Calculations

With some trickery, one can extend these ideas to compute some values of χ_K .

Example 36. Fix a quadratic extension $K/\mathbb{Q}$. Then $\chi_K(-1) \in \{\pm 1\}$, and it equals $+1$ if and only if $\sigma_{-1} \in \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$ fixes K . Now, σ_{-1} is the unique automorphism which sends ζ to $\bar{\zeta}$ for all roots of unity ζ , but this must just be complex conjugation! We conclude that $\chi_K(-1) = +1$ if and only if K is fixed by complex conjugation. Writing $K = \mathbb{Q}(\sqrt{d})$, we see that

$$\chi_{\mathbb{Q}(\sqrt{d})}(-1) = \begin{cases} +1 & \text{if } d > 0, \\ -1 & \text{if } d < 0. \end{cases}$$

Proposition 37. Fix a quadratic extension $K = \mathbb{Q}(\sqrt{d})$, and let p be an odd prime not dividing d . Then

$$\chi_K(p) = \begin{cases} +1 & \text{if } d \pmod{p} \text{ is square,} \\ -1 & \text{if } d \pmod{p} \text{ is non-square.} \end{cases}$$

Proof. By Theorem 24, we may embed K into $\mathbb{Q}(\zeta_N)$. Now, $\chi_K(p) \in \{\pm 1\}$, and $\chi_K(p) = +1$ if and only if $\sigma_p \in \text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$ fixes K , which is equivalent to $\sigma_p(\sqrt{d}) = \sqrt{d}$. Using the embedding $K \subseteq \mathbb{Q}(\zeta_N)$, we may write

$$\sqrt{d} = \sum_{\substack{1 \leq i \leq N \\ \gcd(i, N)=1}} a_i \zeta_N^i$$

for some integers $\{a_i\}$.¹ Then

$$\sigma_p(\sqrt{d}) = \sum_{\substack{1 \leq i \leq N \\ \gcd(i, N)=1}} a_i \zeta_N^{pi}.$$

Now, because p is odd, we may detect if $\sigma_p(\sqrt{d}) = \sqrt{d}$ by passing to $(\text{mod } p)$.

Remark 38. There is something sneaky happening here. We are trying to distinguish $+1$ and -1 in $\mathbb{Z}[\zeta_N]/(p)$, which is equivalent to having $2 \not\equiv 0 \pmod{p}$, which is equivalent to having $2/p \notin \mathbb{Z}[\zeta_N]$. Some studying of algebraic integers is required to guarantee this! For example, $2/p \in \mathbb{Z}[\zeta_N]$ implies that the norm of $2/p$ should be integer, but this norm is $(2/p)^{[\mathbb{Q}(\zeta_N):\mathbb{Q}]}$.

But taken $(\text{mod } p)$, we know $(a+b)^p \equiv a^p + b^p$ for any a and b , so $\sigma_p(\sqrt{d}) \equiv (\sqrt{d})^p$. Thus, $\chi_K(p) = +1$ if and only if $(\sqrt{d})^p \equiv \sqrt{d} \pmod{p}$, which is equivalent to

$$d^{(p-1)/2} \stackrel{?}{\equiv} 1 \pmod{p}.$$

We now do casework on d .

- If d is a square $d = e^2$, then the left-hand side is $e^{p-1} \pmod{p}$, which is $+1$ because $(\mathbb{Z}/p\mathbb{Z})^\times$ has order $(p-1)$.
- If d is not a square, we claim that the left-hand side is not $1 \pmod{p}$. Indeed, the polynomial $x^{(p-1)/2} - 1$ has at most $(p-1)/2$ roots over $\mathbb{F}_p$, and we have already exhibited $(p-1)/2$ roots (from the squares). ■

Example 39. Proposition 37 has interesting applications to squares $(\text{mod } p)$. For example, by Example 34, we see that $2 \pmod{p}$ is a square if and only if $p \equiv \pm 1 \pmod{8}$. This is an instance of quadratic reciprocity!

¹ It is not obvious that the $\{a_i\}$ can be taken to be integers, but it is true. For example, one can see this by unwinding the proof of Proposition 45, writing $\sqrt{d}$ as a product of Gauss sums.

Remark 40. For experts, we remark that there is a proof of Proposition 37 along the lines of Example 36. Note that $\chi_K(p) = +1$ if and only if σ_p fixes K . Thus, we are interested in when the Frobenius Frob_p fixes K . Because p is unramified in K , we may embed $K \subseteq \mathbb{Q}_p^{\text{unr}}$, and then Frob_p (topologically) generates the Galois group $\text{Gal}(\mathbb{Q}_p^{\text{unr}}/\mathbb{Q}_p)$, so the Frobenius fixes K if and only if $K \subseteq \mathbb{Q}_p$. Writing $K = \mathbb{Q}(\sqrt{d})$, this is equivalent to having $\sqrt{d} \in \mathbb{Q}_p$. Because $\sqrt{d}$ is integral, this is equivalent to having $\sqrt{d} \in \mathbb{Z}_p$, which is equivalent to having $d \in \mathbb{F}_p^{\times 2}$ by Hensel's lemma.

3.3 The Conductor

The previous subsection has allowed us to compute some specific values of χ_K , but it is not at all obvious from this calculation what the conductor should be. By Lemma 33, this amounts to computing the minimal N for which $K \subseteq \mathbb{Q}(\zeta_N)$.

It will turn out that the most interesting case is when K is $\mathbb{Q}(\sqrt{\pm p})$ for odd primes p . It turns out to be reasonable to expect that K is contained in $\mathbb{Q}(\zeta_p)$, but because $\text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ is cyclic, this Galois group the field $\mathbb{Q}(\zeta_p)$ admits a unique quadratic sub-extension. So which of $\mathbb{Q}(\sqrt{p})$ or $\mathbb{Q}(\sqrt{-p})$ is in $\mathbb{Q}(\zeta_p)$?

Notation 41. Fix an odd prime p . Then define

$$p^* := \begin{cases} +p & \text{if } +p \equiv 1 \pmod{4}, \\ -p & \text{if } -p \equiv 1 \pmod{4}. \end{cases}$$

Remark 42. By construction, $p^* \equiv 1 \pmod{4}$ for all odd primes p .

Lemma 43. Fix an odd prime p . The unique quadratic sub-extension of $\mathbb{Q}(\zeta_p)/\mathbb{Q}$ is $\mathbb{Q}(\sqrt{p^*})$.

Proof. Let K be a quadratic sub-extension of $\mathbb{Q}(\zeta_p)$. Then $\text{Gal}(\mathbb{Q}(\zeta_p)/K) \subseteq \text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ is a subgroup of index two, but $\text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ is isomorphic to the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$, so there is a unique such subgroup (generated by the square of a generator).

Thus, there is in fact a unique quadratic sub-extension. To generate it, we should find some elements of $\mathbb{Q}(\zeta_p)$ invariant under $\text{Gal}(\mathbb{Q}(\zeta_p)/K)$. Under the identification $\text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$, the previous paragraph tells us that $\text{Gal}(\mathbb{Q}(\zeta_p)/K)$ is given by the squares. Thus,

$$h_p := \sum_{i \in \mathbb{F}_p^{\times 2}} \zeta_p^{i^2}$$

is in K . (In fact, because $\{\sigma \zeta_p\}_\sigma$ is a normal basis of $\mathbb{Q}(\zeta_p)$ over $\mathbb{Q}$, we know a priori that h_p should generate K .) It will turn out to be more convenient to compute the "Gauss sum"

$$g_p := \sum_{i \in \mathbb{F}_p^\times} \chi_p(i) \zeta_p^i,$$

where $\chi_p: \mathbb{F}_p^\times \rightarrow \mathbb{C}^\times$ is the quadratic character of Example 4. An elementary manipulation shows that $2h_p + 1 = g_p$ because $\sum_{i \in \mathbb{F}_p^\times} \zeta_p^i = -1$, so $g_p \in K$ as well (and should generate it). We will calculate g_p in three steps.

1. We claim that $|g_p|^2 = p$. This is an instance of "square root cancellation." By direct expansion, we see that

$$|g_p|^2 = \sum_{i, j \in \mathbb{F}_p^\times} \chi_p(i/j) \zeta_p^{i-j}.$$

We may re-index this sum as

$$|g_p|^2 = \sum_{i,j \in \mathbb{F}_p^\times} \chi_p(i) \zeta_p^{(i-1)j}$$

If $i \neq 1$, then the sum over j is $\sum_j \chi_p(i) \zeta_p^j$ after re-indexing, which is $-\chi_p(i)$. If $i = 1$, then the sum over j is $\sum_j 1 = (p-1)$. Thus, we are done after noticing that $\sum_{i \in \mathbb{F}_p^\times} \chi_p(i) = 0$.

2. We claim that $\bar{g}_p = \chi_p(-1)g_p$. Well,

$$\bar{g}_p = \sum_{i \in \mathbb{F}_p^\times} \chi_p(i) \zeta_p^{-i},$$

so the claim follows from re-indexing.

3. We claim that g_p equals $\sqrt{p^*}$, up to sign. By the previous two points, we see that $g_p^2 = \chi_p(-1)p$, so it remains to show that $\chi_p(-1)p = p^*$. Well, $\chi_p(-1) \in \{\pm 1\}$, and $\chi_p(-1) = +1$ if and only if $-1 \pmod{p}$ is a square. By the proof of Proposition 37, this is equivalent to having $(-1)^{(p-1)/2} \equiv +1 \pmod{p}$, which is equivalent to $(p-1)/2$ being even, which is equivalent to $p \equiv 1 \pmod{4}$. ■

Remark 44. One can use this to show quadratic reciprocity in the following form: let p and q be distinct odd primes. Then we claim that $p^* \pmod{q}$ is a square if and only if $q \pmod{p}$ is a square. Indeed, by Proposition 37, we see that p^* is a square modulo q if and only if $\chi_{\mathbb{Q}(\sqrt{p^*})}(q) = +1$, which is equivalent to the $\sigma_q \in \text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ fixing $\mathbb{Q}(\sqrt{p^*})$ (by Lemma 43), which is equivalent to σ_q living in the unique index-two subgroup of $\text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$, which is equivalent to $q \in (\mathbb{Z}/p\mathbb{Z})^\times$ being a square.

We are now ready to explicate the quadratic sub-extensions of $\mathbb{Q}(\zeta_N)$ for general N .

Proposition 45. Fix a positive integer N . Let $S(N)$ be the set of integers p^* for each prime divisor p of N , along with -1 if $4 \mid N$ and $+2$ if $8 \mid N$. Then for a squarefree integer d has $\mathbb{Q}(\sqrt{d}) \subseteq \mathbb{Q}(\zeta_N)$ if and only if d is a product of elements of $S(N)$.

Proof. The reverse direction is easier to show: by taking products, it is enough to show that $\sqrt{d} \in \mathbb{Q}(\zeta_N)$ for each $d \in S(N)$. Thus, we simply note that $\sqrt{p^*} \in \mathbb{Q}(\zeta_p)$ by Lemma 43, $\sqrt{-1} \in \mathbb{Q}(\zeta_4)$ because $\zeta_4 = \sqrt{-1}$, and $\sqrt{2} \in \mathbb{Q}(\zeta_8)$ because $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$.

For the forward direction, we note that $\mathbb{Q}(\sqrt{d})$ is distinct for each nontrivial squarefree product of elements in $S(N)$, so it is enough to show that there are exactly $2^{\#S(N)} - 1$ quadratic sub-extensions of $\mathbb{Q}(\zeta_N)$. (Thus, the previous paragraph will have exhausted all quadratic sub-extensions of $\mathbb{Q}(\zeta_N)$.) Thus, we would like to show that there are exactly $2^{\#S(N)} - 1$ subgroups of $\text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$ of index two, for which it is equivalent to count subgroups of $(\mathbb{Z}/N\mathbb{Z})^\times$ of index two. Adding in the trivial squarefree product, we would like to show that

$$\# \text{Hom}((\mathbb{Z}/N\mathbb{Z})^\times, \{\pm 1\}) \stackrel{?}{=} 2^{\#S(N)}.$$

By the Chinese remainder theorem, the left-hand side is multiplicative in N . The right-hand side is also multiplicative in N , so it is enough to check this equality on prime-powers.

- If N is an odd prime-power, then $(\mathbb{Z}/N\mathbb{Z})^\times$ is cyclic of even order, so there is exactly one nontrivial homomorphism $(\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \{\pm 1\}$.
- If N is an even prime-power, then we must do some casework. For example, if $N = 2$, then $(\mathbb{Z}/2\mathbb{Z})^\times$ is trivial, so there is only the trivial homomorphism. Similarly, if $N = 4$, then $(\mathbb{Z}/4\mathbb{Z})^\times$ is cyclic of order two, so there is exactly one nontrivial homomorphism. Lastly, if N is divisible by 8, then $(\mathbb{Z}/N\mathbb{Z})^\times \cong \{\pm 1\} \times \mathbb{Z}/(N/4)\mathbb{Z}$,² so there are a total of four homomorphisms to $\{\pm 1\}$. ■

Reversing this argument allows us to compute the conductor χ_K .

² In short, one can show that the elements of $(\mathbb{Z}/N\mathbb{Z})^\times$ which are equivalent to 1 $\pmod{4}$ is a cyclic subgroup generated by 5 $\pmod{N}$.

Corollary 46 (Quadratic reciprocity). Let d be a squarefree integer, and set $K := \mathbb{Q}(\sqrt{d})$. Then the conductor of χ_K is

$$\begin{cases} |d| & \text{if } d \equiv 1 \pmod{4}, \\ 4|d| & \text{otherwise.} \end{cases}$$

Proof. By Lemma 33, the conductor N of χ_K equals the smallest N for which $K \subseteq \mathbb{Q}(\zeta_N)$. We use Proposition 45 to compute this N . Let's start with some easier bounds.

- By Proposition 45, $\sqrt{d} \in \mathbb{Q}(\zeta_N)$ implies that each prime factor p of d must divide N . Because d is squarefree, we see that $d \mid N$.
- Note that $\mathbb{Q}(\zeta_{4|d|})$ contains all needed prime factors of d as well as $\sqrt{-1}$, so d is certainly a product of elements of $S(4d)$, so $\sqrt{d} \in \mathbb{Q}(\zeta_{4|d|})$. Thus, $N \mid 4|d|$.

To refine these bounds, we must do casework on $d \mid 4$.

- Suppose $d \equiv 2 \pmod{4}$. Then by Proposition 45, we must have $2 \in S(N)$, so $8 \mid N$, so comparing prime factors yields $N \geq 4|d|$, and equality follows.
- Suppose $d \equiv 1 \pmod{4}$. Then by Proposition 45, we see that $d = \prod_{p \mid d} p^*$ (recall Remark 42), so $d \in \mathbb{Q}(\zeta_{|d|})$, so $N \leq |d|$, so equality follows.
- Suppose $d \equiv 3 \pmod{4}$. The argument of the previous point yields $\sqrt{-d} \in \mathbb{Q}(\zeta_{|d|})$. Thus, for $\sqrt{d} \in \mathbb{Q}(\zeta_N)$, we see that $\sqrt{-1} \in \mathbb{Q}(\zeta_N)$ as well, so $4 \mid N$ by Proposition 45. We conclude that $N \geq 4|d|$, so equality follows. ■

Remark 47. In light of Proposition 37, we see that one can determine if $d \pmod{p}$ is a square purely based on the information of $p \pmod{4|d|}$. Thus, it is not surprising that quadratic reciprocity is closely related to these ideas! It is possible, though inconvenient, to use Corollary 46 to prove quadratic reciprocity (and some basic facts about quadratic residues) in its usual form. As a sketch, we note that it is more convenient to show quadratic reciprocity for the Kronecker symbol, for which one can induct. Details can be found in Appendix A.

A Quadratic Reciprocity

In this appendix, we explain how to derive the usual statement of quadratic reciprocity from Corollary 46. For reasons relating to induction, it will be more convenient to pass to the Kronecker symbol.

Definition 48 (Kronecker symbol). For any integer d , define the *Kronecker symbol* $\left(\frac{d}{\cdot}\right)$ to be the primitive Dirichlet character associated to $\chi_{\mathbb{Q}(\sqrt{d})}$.

Remark 49. The supplements $\left(\frac{-1}{n}\right) = (-1)^{(n-1)/4}$ and $\left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$ (for odd n) follow from Examples 31 and 34, so we will not worry about them in the sequel.

Lemma 50. Fix an odd positive integer n . Then $\left(\frac{\cdot}{n}\right)$ is a homomorphism $(\mathbb{Z}/n\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$.

Proof. For a given d coprime to n , note that $\left(\frac{d}{\cdot}\right)$ is multiplicative, so

$$\left(\frac{d}{n}\right) = \prod_p \left(\frac{d}{p}\right)^{v_p(n)}.$$

Thus, by the Chinese remainder theorem, it is enough to consider the case where n is an odd prime p . In this case, unwinding the proof of Proposition 37 shows that

$$\left(\frac{d}{p}\right) \stackrel{?}{\equiv} d^{(p-1)/2} \pmod{p}.$$

Indeed, the proof there shows that the left-hand side is $+1$ if and only if the right-hand side is $+1$, but both sides square to $+1$ and hence live in $\{\pm 1\}$, so the equivalence follows. (Here, we are using that p is odd to distinguish $+1$ and -1 after passing to $\pmod{p}$.) The lemma now follows from this equivalence. ■

Remark 51. Even if n is negative, $\left(\frac{\cdot}{n}\right)$ continues to be multiplicative. One just needs to add a $\left(\frac{\cdot}{-1}\right)$ term to the product.

We are now ready to prove quadratic reciprocity.

Notation 52. For any odd integer m , define $m^* := \left(\frac{-1}{m}\right)m$.

Remark 53. Note that $m^* \equiv 1 \pmod{4}$ for all odd integers m . This extends the definition from when m was prime.

Theorem 54 (Quadratic reciprocity). Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right) = \left(\frac{q^*}{p}\right).$$

Proof. We will show the following more general statement: for any coprime odd integers m and n , we have

$$\left(\frac{m^*}{n}\right) \stackrel{?}{=} \left(\frac{n}{|m|}\right).$$

The key input will be that $\left(\frac{d}{-}\right)$ is periodic (mod d) when $d \equiv 1 \pmod{4}$, which is essentially the content of Corollary 46!

Let $Q(m, n)$ be the statement that $\left(\frac{m^*}{n}\right) = \left(\frac{n}{|m|}\right)$ is true for given m and n . We will show $Q(m, n)$ by induction on $|m| + |n|$. Let's deal with signs first.

- Observe that neither side changes if we send $m \mapsto -m$, so $Q(m, n)$ and $Q(-m, n)$ are equivalent.
- Similarly, $Q(m, n)$ and $Q(m, -n)$ are equivalent: we may as well assume $m > 0$ by the previous sentence, and then

$$\left(\frac{m^*}{-n}\right) = \left(\frac{m^*}{-1}\right) \left(\frac{m^*}{n}\right)$$

while

$$\left(\frac{-n}{m}\right) = \left(\frac{-1}{m}\right) \left(\frac{n}{m}\right),$$

so it is enough to note that $\left(\frac{m^*}{-1}\right)$ and $\left(\frac{-1}{m}\right)$ both equal $+1$ if $m \equiv 1 \pmod{4}$ and equal -1 if $m \equiv 3 \pmod{4}$.

This allows us to quickly reduce the base case $|m| + |n| = 2$ to the case $(m, n) = (1, 1)$, for which the given equality reads $1 = 1$.

We now proceed with the induction. By rearranging signs, we may assume m and n are positive. We will have two cases: either $m < n$ or $m > n$.

- Suppose $m < n$. On one hand, Corollary 46 implies

$$\left(\frac{m^*}{n}\right) = \left(\frac{m^*}{n-2m}\right)$$

because $m^* \equiv 1 \pmod{4}$. On the other hand,

$$\left(\frac{n}{m}\right) = \left(\frac{n-2m}{m}\right)$$

by Lemma 50. Thus, $Q(m, n)$ is equivalent to $Q(m, n-2m)$, and the latter holds by induction.

- Suppose $m > n$. On one hand, Lemma 50 implies

$$\begin{aligned} \left(\frac{m^*}{n}\right) &= \left(\frac{\left(\frac{-1}{m}\right)}{n}\right) \left(\frac{m}{n}\right) \\ &= \left(\frac{\left(\frac{-1}{m}\right)}{n}\right) \left(\frac{m-2n}{n}\right) \\ &= \left(\frac{\left(\frac{-1}{m}\right)}{n}\right) \left(\frac{\left(\frac{-1}{m-2n}\right)}{n}\right) \left(\frac{(m-2n)^*}{n}\right). \end{aligned}$$

On the other hand, rearranging with Corollary 46 and Lemma 50 yields

$$\begin{aligned}
 \left(\frac{n}{m}\right) &= \left(\frac{\left(\frac{-1}{n}\right)}{m}\right) \left(\frac{n^*}{m}\right) \\
 &= \left(\frac{\left(\frac{-1}{n}\right)}{m}\right) \left(\frac{n^*}{m-2n}\right) \\
 &= \left(\frac{\left(\frac{-1}{n}\right)}{m}\right) \left(\frac{n^*}{m-2n}\right) \\
 &= \left(\frac{\left(\frac{-1}{n}\right)}{m}\right) \left(\frac{\left(\frac{-1}{n}\right)}{m-2n}\right) \left(\frac{n}{m-2n}\right) \\
 &= \left(\frac{\left(\frac{-1}{n}\right)}{m}\right) \left(\frac{\left(\frac{-1}{n}\right)}{m-2n}\right) \left(\frac{n}{|m-2n|}\right).
 \end{aligned}$$

The last equality holds because $\left(\frac{n}{\pm 1}\right) = 1$ for $n > 0$ by Example 36. We conclude that $Q(m, n)$ is equivalent to $Q(m-2n, n)$ because $\left(\frac{(-1/a)}{b}\right) = (-1)^{(a-1)(b-1)/4}$ for any odd integers a and b . But $Q(m-2n, n)$ holds by induction. ■

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